Netflix Viewing Trends Analysis

Project Overview

The Netflix Viewing Trends Analysis project focuses on analyzing Netflix content data to identify patterns in genres, release trends, and audience preferences across different regions. By leveraging data analytics techniques and interactive dashboards, the project provides insights into how Netflix content evolves over time and which types of content are most popular among viewers.

The goal of this project is to transform raw Netflix dataset information into meaningful insights that can help understand streaming trends and user interests.

Problem Statement

With the rapid growth of streaming platforms, understanding viewer preferences and content distribution trends has become essential. Large datasets containing thousands of movies and TV shows make it difficult to manually analyze patterns. This project aims to use data analytics and visualization techniques to uncover trends in Netflix content.

Technology Stack

Programming Language - Python
Data Analysis - Pandas, NumPy
Data Visualization - Matplotlib, Seaborn
Business Intelligence - Power BI
Data Processing - Exploratory Data Analysis (EDA), Data Cleaning
Tools & Environment - Jupyter Notebook, Google Colab, GitHub

Methodology

1. Data Collection

- The dataset containing Netflix content information was obtained from a public data source.
- The dataset included attributes such as:
    - Title
    - Genre
    - Release Year
    - Country
    - Duration
    - Type (Movie or TV Show)

2. Data Cleaning and Preprocessing

- The raw dataset was cleaned to improve data quality.
- Key preprocessing steps included:
    - Handling missing values
    - Removing duplicate records

    - Standardizing genre and category names
    - Converting date and numeric fields into appropriate formats
- Data cleaning ensured the dataset was accurate and ready for analysis.

3. Exploratory Data Analysis (EDA)

- Exploratory analysis was conducted to understand the structure and characteristics of the dataset.
- Key analysis performed:
    - Distribution of movies vs TV shows
    - Frequency of content release by year
    - Genre popularity analysis
    - Regional content production trends
- Visualization techniques such as bar charts, line charts, and pie charts were used to identify patterns.

4. Feature Analysis and Trend Identification

- Different attributes were analyzed to identify significant trends:
    - Genre popularity trends over time
    - Content production growth by year
    - Country-wise content distribution
    - Comparison between movie and TV show releases
- Statistical summaries and grouped aggregations were used to extract meaningful insights.

5. Data Visualization

- Visualizations were created using Matplotlib and Seaborn in Python.
- Key visualizations included:
    - Genre distribution charts
    - Content release timeline graphs
    - Regional production comparisons
    - Category-wise content analysis

6. Dashboard Development in Power BI

- Cleaned data was imported into Power BI for interactive dashboard creation.
- The dashboard included:
    - Genre popularity charts
    - Content release trends over time

    - Regional distribution of Netflix content
    - Interactive filters for exploring the dataset

7. Insight Generation

- Data visualizations and dashboards were analyzed to derive actionable insights about Netflix content trends.

Key Insights

Popular Genres

- Certain genres such as Drama, Comedy, and Documentary show higher content production compared to niche genres.

Regional Content Distribution

- The analysis revealed that North America and Asia contribute a large share of Netflix content production.

Content Release Trends

- There has been a steady increase in Netflix content production in recent years, reflecting the growth of the streaming industry.